

Jay Patel

Software Engineer · Full-Stack, Mobile and Applied AI

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SUMMARY

Engineer who trains small language models from scratch and ships AI features into production apps. PyTorch training loops, tokenizers, and quantization-aware training on one side; Gemini, RAG, and MCP tool wiring inside client systems on the other.

SKILLS

Machine learning	PyTorch, TensorFlow/Keras, scikit-learn, LSTM, BitNet b1.58, quantization, hyperparameter tuning
Language models	Hugging Face, SentencePiece, RAG, Qdrant, Chroma, MCP, Ollama, LangChain, n8n, Hermes Agent, OpenClaude
MLOps	Docker, FastAPI, Prometheus, CI/CD, model registry (Hugging Face)
Data	Python, Pandas, NumPy, Puppeteer pipelines, BullMQ, MongoDB, PostgreSQL

PROJECTS

- Vini. ~50M BitNet b1.58 edge assistant trained from scratch in PyTorch; pretrained on ~1.05B tokens (FineWeb-Edu, StarCoder, tool traces) on one L4, curriculum-staged instruction data, tuning in progress. Configs: jayptl-rq/vini-pico.
- Tini. ~85M dense TypeScript coding model (RoPE, SwiGLU, GQA) trained from scratch on ~257k deduplicated samples from The Stack; 32k SentencePiece tokenizer, 46h on a T4, val loss 2.07 to 1.31. Published on Hugging Face, Apache 2.0.
- Self-hosted AI agent stack. Local LLM inference, n8n workflows with MCP tool servers, and RAG pipelines over Qdrant and Chroma, running on my own hardware.
- Ai-Vestor, 1st place FinTech, CHARUSAT-Odoo Hackathon 2025. Stock prediction with 100+ LSTM models (90% accuracy) and a gamified investing LMS; FastAPI on Oracle Cloud. 650+ applicants, 75 teams.
- Tenet PG Management. Hostel platform in a Bun Turborepo: Next.js 16 dashboard and Hono API with SSE, UPI payments, WhatsApp Cloud API, PDF invoices, Playwright and Vitest.

EXPERIENCE

Thhiya · Full Stack Developer	Nov 2025 to Jun 2026
- Integrated Google Gemini for provider matching and compliance insights. - Built Puppeteer pipelines that seeded compliance data for 195 countries (payroll, benefits, tax, entities).	
Independent Software Engineer · freelance and contract	Jun 2022 to present
Aviz Health · Full-Stack Developer	Dec 2025 to present
- Architected three Flutter apps and a Next.js 16 admin console; live on Google Play. - Built the Bun + Hono + MongoDB backend with Redis for real-time matching, booking, and SOS dispatch.	
Genuinest · Founding Engineer (Contract)	Jan 2026 to present
Moved the stack off high-cost cloud to Docker, Coolify, Traefik, and Tailscale; self-hosted MongoDB, Redis, RabbitMQ, and a 20TB MinIO cluster, ending recurring storage bills.	
Swalook CRM · Full-Stack Developer	Jan 2026 to present
Bun + Hono + Drizzle backend on PostgreSQL, Redis, and BullMQ; 35+ OpenAPI-documented REST endpoints.	
The Neural Networker · Web Development and React Native Intern	Jun 2025 to Nov 2025
AI-integrated Next.js and React Native interfaces; MongoDB index tuning for 30% faster analytics dashboards.	

EDUCATION AND RECOGNITION

- B.Tech, Information Technology, A.D. Patel Institute of Technology (CVM University), 2022 to 2026
- 1st place, FinTech track, CHARUSAT-Odoo Hackathon 2025 (650+ applicants) · Technical Leadership Award, Apr 2025
- Certifications: Google Cloud Associate Cloud Engineer, AWS Cloud Practitioner, MongoDB Node.js Developer, Meta Front-End Developer, DataCamp SQL Associate